

When Outlier Supervision Fails: Supplementary Material

Anonymous CVPR submission

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Reproducibility Statement

All results are reproducible from the released materials: code (a documented fork of RESfM’s public repository), every training and evaluation configuration file, the rebuilt track suites for the five OOD datasets and the per-domain train/val/test splits of the main paper’s in-distribution study, and the scripts that regenerate every table and figure from the saved evaluation outputs (including a verification script that recomputes each printed cell). Training protocol, learning rates, schedules, and checkpoint-selection rules are specified in the main paper’s protocol section; seeds are enumerated per experiment (20–24), and every headline comparison reports five-training-seed bands. Sources of irreducible run-to-run variance (robust-BA solver nondeterminism, TTT) are characterized in the supplementary so that deviations within those bounds can be recognized as environmental. Dataset provenance and licences: all datasets are public academic benchmarks; our track builder and download manifests are included.

Ethics Statement

This work studies robustness of 3D reconstruction on standard public benchmarks of landmark photographs (MegaDepth, 1DSfM, Strecha, BlendedMVS, Olsson’s dataset) and introduces no new data collection, no personal data processing beyond what those established benchmarks already contain, and no human subjects. Improved robust SfM has broadly beneficial applications (mapping, cultural heritage, robotics); we are not aware of ethical concerns specific to this method beyond those generic to 3D reconstruction of public imagery.

A. Full 1DSfM-hard mechanism table

B. Rotation and registration panels of the main comparison

Rotation errors and registered-camera fractions for every arm of the main comparison (Table 2 of the main paper); conventions as there.

Table 1. 1DSfM-hard ($\sim 60\%$ outliers, seed 20): translation/rotation and registered fraction.

Method	Trans/Rot (post-BA)	Registered
remove	10.6/12.4	all
madweight	12.2/14.4	all
remove+TTT	15.2/16.8	all
RESfM-scratch	16.2/22.7	all
huberweight	16.9/24.5	all
COLMAP	20.9/–	64%
weight	21.9/25.6	all
stdweight	23.2/27.3	all
hybrid	26.3/28.7	all
weight+TTT	26.4/32.0	all

Arm	MegaDepth	1DSfM	1DSfM-hard	Strecha	BMVS	Olsson
Rotation error ($^\circ$, five-seed means; classical row single run)						
madweight	3.70	7.74	27.7	12.80	9.01	14.4
weight	3.28	8.24	28.0	7.61	9.17	13.2
weight+TTT	2.88	10.09	30.6	7.95	8.30	11.6
remove	4.04	6.94	12.3	14.39	16.04	13.9
remove+TTT	4.22	6.90	16.4	13.23	13.19	14.5
GLOMAP (classical)	5.30	8.78	26.4	0.25	2.15	1.29
RESfM-scratch (supervised)	1.89	9.62	27.3	1.56	31.66	12.6
Registered cameras N_r/N_c (five-seed means; classical row single run)						
madweight	81%	70%	43%	91%	92%	86%
weight	79%	79%	40%	98%	94%	90%
weight+TTT	77%	81%	41%	99%	94%	90%
remove	72%	58%	29%	97%	88%	78%
remove+TTT	71%	58%	32%	96%	90%	78%
RESfM-scratch (supervised)	85%	77%	34%	99%	92%	89%
GLOMAP (classical)	96%	96%	80%	100%	86%	100%

C. Training-pool diversity and the label ceiling

A preliminary experiment (single seed, under a coarser five-milestone decay schedule; cf. the main paper’s protocol section) retrains both methods on progressively diversified pools: MegaDepth-27 $\rightarrow$ +ETH3D (39 scenes, added data $\sim 8\%$ contaminated) $\rightarrow$ +VGG/T&T (52 scenes, adding a $\sim 16\%$ -contaminated mid-band). The supervised baseline’s clean-OOD accuracy degrades monotonically with diversity (Strecha 0.006 $\rightarrow$ 0.77 $\rightarrow$ 2.83); in distribution the same diversification is nearly free (MegaDepth 0.33 $\rightarrow$ 0.32 $\rightarrow$ 0.44, the first step within noise), so the collapse is specific to clean OOD generalization, not a general capacity cost. Meanwhile the clean addition yielded no high-

Roman Forum (60% contamination), both trained in-domain: BA-aligned camera centers, top view



Figure 1. Qualitative view of the 60% label ceiling (Roman Forum, seed 20, both recipes trained in-domain, supervised with GT-derived labels): BA-aligned camera centers in top view; whiskers join each reconstructed camera to its ground-truth position. The label-free madweight arm (left) registers more cameras (67 vs. 54 of 300) that track the ground-truth layout at a usable median error (3.8 vs. 15.4), while the supervised reconstruction (right) collapses toward a compressed cluster with large systematic displacements — supervision fails here with the best labels it could ever have.

contamination gain and only the contaminated mid-band coincided with one (1DSfM 11.0/11.5/8.8; differences within single-seed noise on this benchmark, so consistent with, though not established by, these runs). If it holds, the implication is pointed: the data that helps is precisely the data whose COLMAP labels degrade (cf. the label ceiling), making supervised diversity scaling self-limiting.

A four-cell control (single seed each, canonical schedule) sharpens the architecture question: on the Olsson clean benchmark, the supervised collapse persists for a deep network on the MegaDepth-only pool (8.58) and for a shallow network on the diversified 52-scene pool (9.30), yet vanishes for deep networks on the diversified pool (2.26 plain; 2.83 with Layer-Norm/residual — so normalization is not the ingredient). Neither capacity nor data diversity alone rescues the supervised recipe; only their combination does. The label-free arms reach the same level (2.9–3.6) from the shallow, single-pool configuration — the cheapest cell of the grid — which reframes the comparison: self-supervision buys at 1×3 capacity on 27 scenes what supervision needs both depth and a curated multi-dataset pool to approach.

A two-stage control sharpens what kind of failure the collapse is. Retraining the supervised recipe from scratch on tracks pre-cleaned by its own best outlier head (with label-free reprojection-based checkpoint selection) does nothing at the full 20k+20k budget — all five seeds collapse (8.3–12.0) — but a compute-matched 10k+10k version escapes in three of five seeds (1.69/2.01/2.29, matching the best cells of the grid) while the other two collapse (7.8/12.1), with the same bimodality on Strecha (0.01–0.2 vs. 1.9/2.4). The good

basin is therefore reachable from the shallow single-pool configuration — the collapse is basin selection, not a capacity or data impossibility — but no supervised-lineage recipe we tested reaches it reliably: across seeds, clean-OOD reliability is delivered only by the label-free arms and by depth $\times$ diversity. Nor do the two remedies compose: applying the same two-stage cleaning on top of the depth $\times$ diversity configuration destroys its fix (Olsson 9.94 vs. 2.83 one-stage) — pre-cleaning strips the very training signal the diverse pool provides — while incidentally producing the best BlendedMVS result in the study (0.10). Data surgery and data diversity are substitute-seeking interventions that interfere, not stack. The label-free arms diverge by mechanism, and the canonical-schedule three-seed replication (authors’ configuration, 52 scenes) now separates the robust from the fragile: madweight’s clean-OOD diversity-robustness is seed-stable (BlendedMVS 0.12 ± 0.01 vs. the baseline’s 0.35 ± 0.04), whereas its high-contamination gain is real only in the best case: 1DSfM 5.4/11.2/12.8 across seeds (9.8 ± 3.9), overlapping the single-pool band (11.5 ± 2.3). Diversification can yield large gains at 43% but does not do so reliably, and yields none at 60% (20.1 ± 1.2); soft weighting collapses on clean OOD like the baseline at the retired schedule (2.9–4.0) and holds only BlendedMVS under the canonical one.

Schedule robustness (from main protocol)

Schedule robustness. All reported results use RESfM’s released eleven-milestone schedule. An earlier accidental five-milestone variant of ours was retired; a three-seed replication of both schedules shows the schedule effect lies within seed noise (MegaDepth 0.46 ± 0.08 vs. 0.40 ± 0.09 ; 1DSfM 9.6 ± 3.2 vs. 12.2 ± 2.4 , per-seed rankings inverting on both), with mechanism orderings stable throughout.

D. Qualitative results

Fig. 2 overlays BA-aligned camera centers on ground truth for 1DSfM NYC_Library. Label-free madweight (2.94) leaves far fewer gross-drift cameras than the supervised from-scratch baseline (6.28), which shows more and larger gross drifts.

E. Learning-rate robustness of the baseline

Our protocol gives each method its best learning rate at the canonical 20k budget (10^{-3} for the supervised arms, RESfM’s released value; 10^{-4} for the self-supervised arms). To test how much the conclusions depend on this choice we trained the supervised baseline at 10^{-4} as well. In distribution the released value

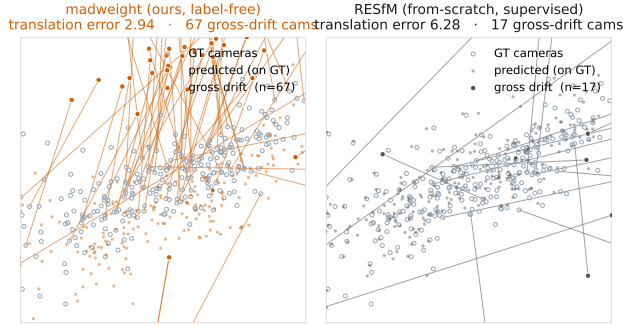


Figure 2. Qualitative 1DSfM (NYC_Library, 32.6% outliers). BA-aligned camera centers vs. GT (open markers); large dots + connectors mark gross-drift cameras. madweight (left, 2.94) has fewer and shorter drifts than RESfM-from-scratch (right, 6.28).

is clearly right (0.344 vs. 0.533), and at 10^{-4} the supervised model is truncation-limited (best checkpoint at the final epoch). Out of distribution, however, the picture splits: the baseline’s Olsson collapse persists at both rates (8.47 vs. 8.54), i.e. the clean-benchmark supervised failure is lr-robust, whereas its Blended-MVS, 1DSfM, and 1DSfM-hard deficits shrink substantially at the lower rate ($0.379 \rightarrow 0.139$, $11.0 \rightarrow 5.9$, $16.2 \rightarrow 12.8$): the under-trained low-rate model overfits MegaDepth less and generalizes better. Two follow-ups complete the grid. A 30k-budget replicate at 10^{-4} plateaus (best validation at epoch 16k, never improving through 30k), and its evaluation sharpens the picture: that checkpoint reaches near-parity in distribution (0.365 vs. the 10^{-3} baseline’s 0.344) while keeping the OOD gains (1DSfM 6.1, 1DSfM-hard 13.9, BlendedMVS 0.18), making it the strongest supervised variant out of distribution (single seed). The Olsson (8.27) and Strecha (1.64) failures persist, and label-free plain removal still leads at extreme contamination (12.4 ± 2.1 vs. 13.9); the OOD advantage of the low rate is a regularization effect, not an under-training artifact. And the self-supervised arms show no symmetric effect: trained at 10^{-3} they are worse on every dataset (madweight MegaDepth 0.75 vs. 0.42, 1DSfM 15.7 vs. 12.4, Olsson 10.1 vs. 2.6). Each method’s declared rate is therefore its empirical best-of-two, and the off-rate OOD gain is specific to supervision, consistent with a COLMAP-label-fitted classifier overfitting in distribution. We scope our claims accordingly: the Olsson and Strecha findings and the mechanism flip itself are lr-robust, while the magnitude of the supervised OOD deficit at BlendedMVS and high contamination depends on the baseline’s learning rate.

Table 2. Full shallow mechanism matrix (RESfM-aligned schedule, seed 20; $\dagger$ mean $\pm$ std over seeds 20–22). Mean / median translation (post-BA); Olsson column: means. Best per column in bold.

Arm	MD	1DSfM	Strecha	BMVS	Olsson
remove	0.60/0.19	$9.1 \pm 2.8 / \mathbf{1.6 \pm 0.2}^\dagger$	3.02/1.68	0.27/0.19	2.81
remove+TTT	0.49/0.13	$\mathbf{8.7 \pm 1.1} / 2.2 \pm 1.1^\dagger$	2.97/1.56	0.19/0.07	4.07
hybrid	0.37/0.20	$16.0 \pm 1.4 / 6.7 \pm 4.4^\dagger$	2.70/1.39	0.10/0.03	2.82
hybrid+TTT	0.39/0.14	$15.4 \pm 1.1 / 10.7 \pm 3.5^\dagger$	3.52/2.65	0.13/0.00	3.19
madweight	0.42/0.15	$11.5 \pm 2.3 / 6.0 \pm 1.4^\dagger$	2.50/0.97	0.15/0.004	2.57
weight	0.42/0.18	$15.8 \pm 3.1 / 10.9 \pm 2.1^\dagger$	2.00/0.03	0.10 /0.04	2.65
weight+TTT	0.39/0.14	$17.0 \pm 1.3 / 10.9 \pm 4.1^\dagger$	1.98/ 0.005	0.14/0.01	7.74
RESfM (sup.)	0.34/0.07	11.03/2.93	0.006/0.005	0.38/0.35	8.47

Table 3. Removal-statistic ablation (head reweighting fixed). Only MAD’s median is seed-stable.

Statistic	1DSfM mean	1DSfM med. s20/21/22	BMVS
MAD (madweight)	11.5 ± 2.3	4.4 / 7.0 / 6.6	0.15
Huber	11.5 $\pm$ 1.6	3.8 / 12.4 / 5.3	0.11
STD	15.2 ± 1.2	13.3 / 9.0 / 12.9	0.15

F. Full mechanism matrix

Table 2 completes the mechanism study (all arms, RESfM-aligned schedule, seed 20). Notably, at 43% the plain-removal family leads every arm including the supervised baseline, and the full three-seed band confirms it: remove+TTT 8.7 ± 1.1 mean (every seed below the baseline’s 11.0), plain remove median 1.59 ± 0.18 (vs. 2.9; means 9.1 ± 2.8 , dominated by the two failure scenes). On clean data the soft arms dominate the removal family, reproducing the complementarity. On outlier-free Olsson the grid flattens (all label-free arms 2.6–4.1 vs. supervised 8.5): the clean-benchmark failure is specific to the supervised head, not the mechanism.

G. Which robust statistic? MAD vs. STD vs. Huber

madweight removes by a robust statistic of the reprojection error; we ablate the statistic holding the head reweighting fixed: MAD (median + $\alpha \cdot 1.4826$ MAD, 50% breakdown), a Huber M-estimate of scale (IRLS), and non-robust STD (mean + $\alpha \sigma$), all at $\alpha=2$. Table 3 reports the ablation over seeds 20–22: only MAD’s 1DSfM median is stable; Huber’s collapses on one seed, and non-robust STD is consistently poor, its mean/ σ threshold inflated by the very outliers it should catch. On clean data all three are comparable: little is removed either way, the right behavior without outliers. The 50%-breakdown statistic is the only reliably robust choice, and we use MAD in madweight.

Table 4. Hyperparameter sensitivity (seed 20, single runs; earlier five-milestone schedule for the percentile variants). Left: pseudo-label percentiles. Right: removal threshold τ on the default checkpoint. Mean translation.

Percentiles	MD (weight)	1DSfM (madweight)	τ	MD	1DSfM
10/90	0.49	13.0	0.4	0.58	8.0
15/85	0.51	15.6	0.5	0.40	10.0
20/80*	0.42	12.4	0.6*	0.39	9.0
25/75	0.60	14.9	0.7	0.38	9.0
30/70	0.44	8.8	0.8	0.43	10.0

H. Sensitivity of the self-supervision hyperparameters

SS-RESfM introduces two hyperparameters: the pseudo-label percentiles (default 20/80) and the removal threshold ($\tau=0.6$, RESfM’s released value). Both were fixed a priori, not tuned per dataset; this appendix reports their sensitivity.

Removal threshold. Sweeping $\tau \in \{0.4, \dots, 0.8\}$ on the default-percentile checkpoint (seed 20): MegaDepth varies only 0.38–0.58 (flat within 0.38–0.43 for $\tau \geq 0.5$), Strecha 3.06–3.22, BlendedMVS 0.20–0.27, and 1DSfM 7.97–10.35 with no consistent optimum; the default $\tau=0.6$ is never far from the per-dataset best. The threshold is a non-issue.

Pseudo-label percentiles. Across a $4\times$ range of percentile choices (Table 4), in-distribution accuracy is flat (MegaDepth 0.42–0.60), and 1DSfM varies by ± 3 –6 — an amount comparable to the training-seed bands of Table 2 of the main paper (± 1.7 –3.0), with no setting consistently best across mechanisms (these are single-seed cells, so part of the spread is seed noise). The default 20/80 sits mid-range everywhere and was chosen before any evaluation. We conclude the method is not sensitively tuned: neither knob, varied broadly, moves results beyond the seed-level noise this paper quantifies everywhere else.

I. Per-scene complementarity and the limits of contamination-based selection

Fig. 3 tests the flip per scene: each scene’s outlier rate vs. the signed advantage of the best removal mechanism over soft weighting. On OOD data, weight wins the low-contamination scenes and removal the high-contamination ones; the crossover is bracketed to the sparsely-sampled ~ 3 –29% band.

Does per-scene contamination help selection? No: within each dataset contamination is too homogeneous to trigger different choices, so a contamination-keyed selector (madweight if scene rate $\geq 10\%$, else weight) collapses to per-dataset selection (Table 5). An or-

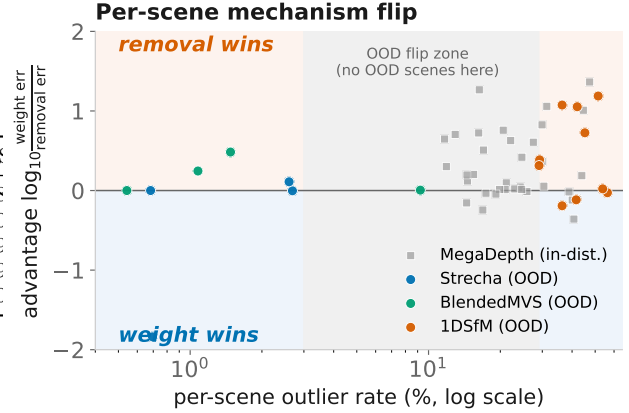


Figure 3. Per-scene complementarity (RESfM-aligned schedule). Each point is one scene: outlier rate (x, log) vs. signed advantage of removal over soft weighting. Weight wins at low contamination, removal at high contamination.

acle choosing the best of five mechanisms per scene from ground-truth error would change the picture: MegaDepth 0.42 $\rightarrow$ 0.116, surpassing every fixed arm, the supervised baseline at 0.344, and even RESfM-released at 0.209, and 1DSfM 12.4 $\rightarrow$ 5.8 vs. the baseline’s 11.0. The performance is thus already latent in the label-free mechanism family; selection, not mechanism design, is the bottleneck. A first unsupervised step exists: running all five mechanisms and keeping the reconstruction with the lowest post-BA reprojection error reaches 0.276 on MegaDepth, past every fixed arm and the supervised baseline (0.344), and matches the oracle on BlendedMVS (Table 5, self-sel.). We report this as a secondary observation rather than a claim: it spends $5\times$ test-time compute, and it inherits reprojection’s retention bias exactly where pruning over-fires (1DSfM 12.8 vs. plain remove’s 8.9). Cheaper selection signals fail informatively. The classification head cannot estimate contamination at all: trained on per-scene percentile pseudo-labels, it is a rank calibrator, not a rate estimator, flagging a near-constant 39–42% of observations on datasets whose true contamination spans 0.5–60% (this also explains the collapse of contamination-keyed selection, and retroactively justifies madweight’s design choice of thresholding reprojection statistics rather than head scores). And pre-fine-tune reprojection magnitudes measure domain shift, not contamination (in-distribution MegaDepth shows the largest raw error); only the post-fine-tune residual tracks the contamination axis monotonically. Realizing the remaining headroom at unit compute is future work.

Table 5. Per-scene selection (mean translation; RESfM-aligned schedule). Contamination-keyed selection collapses to per-dataset choice. The oracle uses ground-truth error over five mechanisms: an upper bound, not a deployable method. Self-sel.: run all five, keep the lowest post-BA reprojection error (unsupervised, $5\times$ test-time compute); a secondary observation, not a headline method.

Dataset	weight	madwt	contam	self-sel. ($5\times$)	oracle (GT)
MegaDepth	0.423	0.422	0.422	0.276	<i>0.116</i>
1DSfM	12.79	12.39	12.39	12.83	<i>5.78</i>
Strecha	2.00	2.50	2.00	2.70	<i>1.98</i>
BlendedMVS	0.100	0.146	0.100	0.083	<i>0.083</i>

J. Environment sensitivity and reproducibility of post-BA benchmarks

This appendix reproduces the full study behind the protocol’s environment-noise and stack-replication statements (main paper, protocol section).

J.1. The reproduction question

Evaluating the RESfM authors’ released checkpoint in our harness on the 36 MegaDepth test scenes yields a post-BA mean translation error of 0.209 vs. the published 0.169 (rotation 1.89° vs. 1.29°). Since our controlled comparisons hinge on baseline fidelity, we traced this discrepancy to its source. The finding is of independent interest: dataset-level post-BA means on MegaDepth-scale benchmarks carry a practical environment-noise floor of $\mathcal{O}(0.05)$ for non-TTT evaluations, produced by chaotic basin selection in robust bundle adjustment — and the floor is larger elsewhere: per-scene values flip by up to ± 0.37 (released checkpoint: $0.043 \rightarrow 0.892$ on scene 0046), TTT-based cells carry ~ 0.14 mean-level run-to-run variance (§J.4), and on high-error benchmarks the floor scales with the error (released checkpoint on 1DSfM: ± 0.27 across runs). Third-party corroboration: the VGPA appendix [?] reports that re-running RESfM under their own protocol shifts its numbers because “the final bundle-adjustment stage is not fully deterministic” — the same environment-noise layer, observed independently by the original authors’ group. This corroborates the noise floor only: the reproduction gap itself (0.169 vs. 0.34–0.39, roughly $4\times$ the floor) traces to training (checkpoint provenance and training-seed variance), not to bundle adjustment.

Table 6. Released RESfM checkpoint, MegaDepth 36-scene post-BA means, under three conditions, vs. the published number. N_r is the mean number of registered cameras per scene; the published row averages the paper’s per-scene Table 1 values.

Condition	Trans	Rot ($^\circ$)	N_r
Published (paper Table 1)	0.169	1.29	298
Our env (pip/OpenBLAS), 5-seed mean	0.203	1.82	304
Our env, seed 20	0.209	1.89	304
Authors’ exact stack (conda/MKL)	0.217	1.92	304

J.2. Eliminating the controllable factors

Data. Our MegaDepth npz files are byte-identical to the authors’ released archive, including the Group-2 300-camera subsamples.

Randomness. Five evaluation seeds vary the pre-BA network output by ± 0.004 and the post-BA mean by ± 0.01 ; negligible.

Network forward pass. The pre-BA output (translation/rotation $\sim 1.73/11.9^\circ$) is essentially seed- and environment-stable; the divergence therefore enters at the robust-BA stage. (Notably, the paper does not report pre-BA numbers; all its tables are post-robust-BA.)

Software stack. We rebuilt the authors’ exact pinned environment from their released environment.yaml: Python 3.8.20, MKL-backed numpy 1.24.3 (mkl=2023.1.0), pycolmap 3.10.0, pyceres 2.3, lightning 2.3.3. (The sole deviation is the CUDA build of PyTorch, forced by GPU-driver compatibility; the BA stage is CPU-only.) Re-evaluating all 36 scenes in this matched environment gives translation/rotation errors of $0.217/1.92^\circ$, statistically indistinguishable from our environment’s $0.209/1.89^\circ$, and not the published $0.169/1.29^\circ$. Registration is equally stable: all three of our conditions register 304 cameras per scene on average (Table 6), slightly more than the published 298, so the higher mean error is not an artifact of evaluating fewer cameras.

J.3. Mechanism: chaotic basin selection in robust BA

Initialization perturbation is at the 10^{-10} level. We isolated the BLAS layer by running the identical DLT triangulation (the BA initialization) on identical inputs under OpenBLAS and MKL, single-threaded: over 11,131 triangulated points the maximum relative difference is 4.8×10^{-10} (mean 5×10^{-14} ; NaN patterns identical).

Yet outcomes move by $\mathcal{O}(0.1)$. This sub-nanoscale perturbation changes the per-scene post-BA translation by up to 0.167, with 16/36 scenes moving by more than 0.01. Robust BA on ill-conditioned scenes amplifies last-bit differences into different convergence basins.

The published gap has the same signature. Comparing our released-checkpoint evaluation to the paper’s Table 1 per scene: 20/36 scenes agree within 0.05 (8 essentially exactly), 17 are better in our runs, and the entire mean gap is carried by four basin flips (scenes 0046, 0148, 0012, 5016; e.g. 0046: $0.043 \rightarrow 0.892$). Bidirectional, scene-concentrated flips are the signature of basin selection, not of a weaker checkpoint, which would shift errors in one direction rather than improve 17 of 36 scenes.

J.4. Stack replication and TTT run-to-run variance

Beyond the released checkpoint, we re-evaluated both from-scratch main-comparison checkpoints under the authors’ pinned stack (MKL numpy, pycolmap 3.10.0, pyceres 2.3), changing nothing else, and additionally repeated the TTT evaluation within our own stack.

The supervised baseline is essentially unchanged across stacks: MegaDepth 0.344 $\rightarrow$ 0.350 mean, 0.070 $\rightarrow$ 0.088 median, only 7/36 scenes moving more than 0.05 (max ± 0.37 , bidirectional). This matches the released-checkpoint result above. The SS-RESfM weight+TTT arm, by contrast, varies regardless of stack: three evaluations of the same checkpoint (ours, a same-stack repeat, the authors' stack) give MegaDepth means 0.316, 0.401, 0.453 (~ 18 –20/36 scenes shifting per repeat, up to ± 2.9 , largely the same scenes, medians 0.12–0.19). The attribution is therefore not the software stack but test-time fine-tuning itself: it runs an optimization at evaluation time, so the chaotic basin selection documented above re-enters the result on every run. Consequently the paper reports all results from the single declared pipeline, treats TTT means as carrying $\sim \pm 0.07$ run-to-run variance, and rests its environment- and run-independent claims on the deterministic no-TTT cells (which reproduce exactly on repeat).

J.5. Are seed replicates necessary? Testing seed-additivity

A natural economy argument runs: if the training seed shifted all models by a shared offset (no seed $\times$ model interaction), relative differences would be seed-invariant and a single seed would rank models correctly. We test this directly on the saved per-scene results: for the pairs (supervised – madweight) and (weight – madweight), we form the per-scene, per-seed difference over the five training seeds across all six datasets (98 scene-cells) and ask whether its sign and magnitude are stable.

They are not. At scene level, the sign of the supervised-vs-madweight difference flips across seeds in 67% of scenes (66/98; per-dataset 50–80%), and the median across-seed standard deviation of the difference matches or exceeds the median margin itself (MegaDepth 0.20 vs. 0.10; 1DSfM 4.5 vs. 3.4) — one to two orders of magnitude above the evaluation-seed and environment-noise floors of §J.2, so this is genuine training-seed interaction, not measurement noise. At dataset level, the supervised-vs-madweight verdict itself flips across seeds on four of six datasets: MegaDepth (–0.34 to +0.10), 1DSfM (–2.9 to +3.6), 1DSfM-hard (–3.1 to +4.0), and Olsson (+5.9 on four seeds, –0.5 on the escape seed); only Strecha (supervised wins every seed) and BlendedMVS (label-free wins every seed) are seed-invariant — and those are exactly the two cells the paper states as every-seed claims. The within-family pair (weight – madweight) flips only where the two arms are near-tied ($|\text{margin}| \leq 0.3$ on MegaDepth, BlendedMVS, Olsson — flips around zero, i.e. genuine ties), with one true instability at 60% con-

tamination, where soft weighting's dataset-level margin swings from –2.3 to +9.7.

The conclusion is the methodological premise of this paper: seed-additivity fails in this benchmark, a single-seed comparison can reverse the headline verdict on four of six datasets, and any cross-recipe claim must therefore be established at the level of seed bands (or, for the strongest claims, all seed pairings), as done throughout.

Couldn't one just pick a good seed? A natural rejoinder is that a single-seed comparison is fine if every method reports a well-chosen seed. Four facts close this off. (1) It is not the protocol in use: RESfM fixes one shared seed (20) for itself and for the baselines it retrains, with no selection step — a single lottery draw, and under the interaction above, a lottery whose variance differs by method. (2) Choosing a good seed requires knowing which seed is good, and for the OOD regimes where the failures live, nothing available at training time identifies it: label-free proxy selectors fail to find the escape seed (the per-scene selection appendix), and the authors' own Accuracy-based checkpoint selection retains the Olsson collapse (8.8). Selecting the seed that avoids a clean-benchmark failure requires evaluating on the clean benchmark itself — test-set selection. (3) The paper's strongest claims are constructed to survive even adversarially favorable seed choices: at 60%, plain remove's worst seed beats the supervised baseline's best (15.8 vs. 16.2), so no seed policy on either side reverses it; the one claim a lucky seed softens (the Olsson margin, where the escape seed ties the label-free band) is stated as a reliability claim — collapse in 4/5 seeds — for precisely this reason. (4) Empirically, the fixed seed was not a lucky one anyway: at seed 20 the baseline is mid-band on MegaDepth and a collapse seed on Olsson, and the released checkpoint (0.209) is itself a favorable draw that the authors' own training code does not reproduce (0.34–0.39 on our tracks) — consistent with published single-seed numbers being draws from a wide distribution rather than representative values.

Classical baselines and run-to-run variance. The same replicate protocol applied to the classical rows sharpens the tiering: GLOMAP is deterministic in practice (five independent runs per dataset; run-to-run std ≤ 0.06 translation everywhere, e.g. MegaDepth 3.33 ± 0.03 , 1DSfM 27.21 ± 0.06), so its single-run rows are representative; incremental COLMAP is not (20.4 ± 1.1 on 1DSfM-hard over five runs, with per-scene registration swinging by up to 67 cameras) — RANSAC-driven incremental reconstruction inherits real run-to-run vari-

ance. The volatility hierarchy is thus: classical global (none) < classical incremental (moderate) < learned recipes (dominant, and concentrated in the supervised arm). A full COLMAP grid (five runs per scene on every dataset) adds nuance: its run-to-run variance is itself contamination-dependent — bit-stable on the clean sets (Olsson 0.20 ± 0.00 , Strecha 0.03, Blended-MVS 0.01, where it is the strongest method in the table) and genuinely variable only at 60% (20.4 ± 1.1). Incremental reconstruction is excellent and reliable exactly where outliers are absent, and fails where they dominate — the classical mirror of the complementarity finding. (COLMAP errors are computed over its registered cameras only, a favorable convention; its two largest 1DSfM scenes are excluded pending completion.)

Even oracle-clean tracks do not confer reliability. ESfM* — the architecture trained on ground-truth-cleaned tracks, with no outlier mechanism in the loop at all — is itself seed-bimodal on BlendedMVS: five training seeds give [0.00, 0.33, 4.93, 5.64, 8.08] (3.79 ± 3.51). The clean-ODD reliability failure therefore lives in the geometric trunk’s training dynamics, not in outlier handling or label quality, and no amount of label perfection removes it — only the adaptive label-free arms happen to escape it in our grid. The completed Olsson column makes the same point without bimodality: ESfM* on oracle-cleaned tracks lands at 9.36 ± 1.28 — inside the supervised collapse band (7.4 ± 2.6) and roughly three times the adaptive arms’ ~ 3.0 on contaminated tracks — so the clean-benchmark collapse survives even ground-truth track cleaning. The one dataset where oracle-clean tracks do win is Strecha (1.14 ± 0.45 vs. the best label-free arm’s 1.97) — the smallest benchmark (54 cameras) and the same one where supervision wins from every training pool (Appendix N); clean tracks can work there, but nowhere that the collapse or the bimodality lives.

J.6. Implications for reading the paper’s numbers

(1) The published number and ours are draws from a distribution over environments; no software matching can force a particular draw. (2) Differences of $\lesssim 0.05$ in post-BA translation means on these benchmarks are within environment noise and should not be read as method differences, by any paper. (3) Consequently, meaningful comparisons must be environment-controlled: every number in the main paper (all arms, RESfM-from-scratch, and the released checkpoint) is produced in a single environment, and the out-of-distribution margins ($2\times$ and larger) sit far above this noise floor.

K. The same results through count- and coverage-based lenses

Our main tables follow the field’s convention (RESfM, ESfM, GASfM): translation/rotation means over each method’s registered cameras, with N_r reported alongside. That convention answers “how accurate is what you built” but not “how much of the scene did you solve correctly,” and it treats a dataset mean as the summary of a distribution that, as the dtu106 decomposition shows, can be tail-dominated. This appendix re-reads the complete five-seed matrix through three complementary lenses: pairwise pose AUC with and without coverage pricing, per-scene win counts, and a catastrophic-cell census. The lenses disagree with the means in instructive places; we report all of them so no single convention carries a verdict alone.

Metrics and recovery. For every (method, dataset, seed, scene) cell we recover the final post-BA cameras from the saved COLMAP reconstructions, identify each camera against the ground truth by its exact intrinsics (falling back to track-coordinate voting where a scene shares intrinsics; identification 97–100% per scene, validated against exact-intrinsics-unique cameras), and compute pairwise relative-pose errors, which are invariant to each reconstruction’s global similarity. AUC@30° averages $(30 - e)^+/30$ over registered pairs (pose error $e = \max$ of relative rotation angle and translation-direction angle); registration-aware RA-AUC@30° averages over all $\binom{N_c}{2}$ pairs, scoring pairs with an unregistered camera at 180° — the coverage-priced convention of the concurrent VGPA work [?].

The coverage-pricing inversion. At 60% contamination the two metric families give opposite orderings. Plain remove — the winner on translation means (12.4 ± 2.1 vs. supervised 19.2 ± 3.4), medians, and win counts (80/84% of cells) — is last under RA-AUC (0.036), because its 29% registration poisons over 90% of camera pairs; the no-mechanism baseline, which loses every raw-error comparison there, tops the RA column (0.099) by registering 40%. AUC over registered pairs shows removal’s accuracy is real (tied-best 0.411): the win is genuine on what it reconstructs, and is bought with coverage. Neither convention is “the” answer: raw error over registered cameras rewards aggressive removal; coverage-priced AUC rewards registering more cameras at moderate error — the in-distribution analogue of the COLMAP registered-only caveat in our main table. Every removal-family claim in this paper should be read with its N_r , and we report

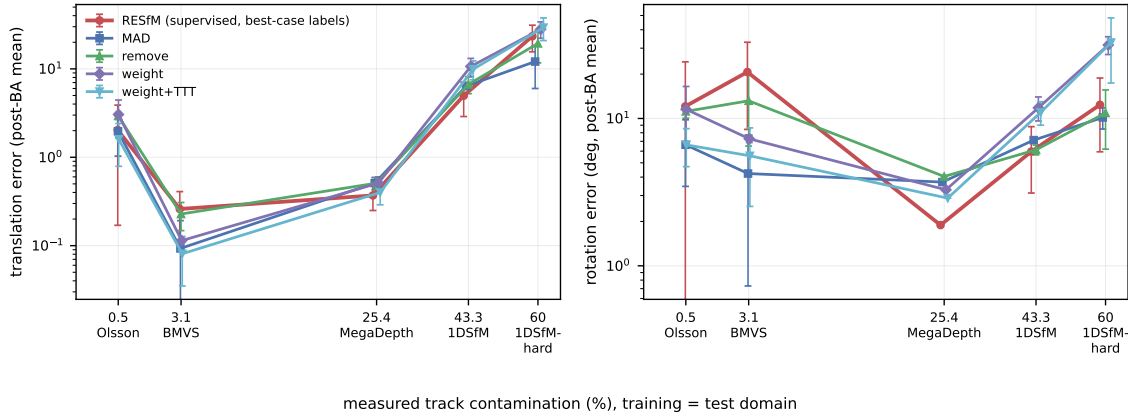


Figure 4. The in-distribution contamination curve. Post-BA mean translation (log scale) when training and test share the domain, versus measured contamination; supervision uses best-case labels. Each label-free arm is its own line: soft weighting ($\pm$ TTT) leads at low contamination, robust removal at high — the mechanism flip, replicated in-distribution. Supervision wins at 25% and narrowly at 43%, is seed-bimodal at 0.5%, and loses at 3.1% (dead zone) and 60% (label ceiling): two distinct failure modes, not a smooth function of contamination. The rotation panel makes the BlendedMVS dead zone starker ($20.7 \pm 12.2^\circ$ vs. $4\text{--}13^\circ$). Bands: $\pm$ std over five seeds.

Table 7. Pose AUC@30°, five-seed mean $\pm$ std. Top block: RA-AUC (coverage-priced). Bottom block: AUC over registered pairs only. Best per column in bold. The two blocks disagree exactly where accuracy is bought with coverage.

	MegaDepth	1DSfM	1DSfM-hard
RA-AUC@30° (all pairs; unregistered = 180°)			
madweight	0.599 $\pm$ 0.02	0.374 $\pm$ 0.01	0.067 $\pm$ 0.01
weight	0.582 $\pm$ 0.02	0.444 $\pm$ 0.02	0.092 $\pm$ 0.01
weight+TTT	0.580 $\pm$ 0.02	0.445 $\pm$ 0.02	0.092 $\pm$ 0.01
remove	0.442 $\pm$ 0.01	0.259 $\pm$ 0.01	0.036 $\pm$ 0.01
remove+TTT	0.442 $\pm$ 0.01	0.262 $\pm$ 0.01	0.037 $\pm$ 0.01
hybrid	0.559 $\pm$ 0.03	0.442 $\pm$ 0.02	0.095 $\pm$ 0.01
RESfM-scratch (sup.)	0.605 $\pm$ 0.01	0.408 $\pm$ 0.01	0.044 $\pm$ 0.01
ESfM (no mech.)	0.574 $\pm$ 0.02	0.461 $\pm$ 0.01	0.099 $\pm$ 0.01
ESfM* (clean tracks)	0.691 $\pm$ 0.04	0.461 $\pm$ 0.02	0.035 $\pm$ 0.01
AUC@30° (registered pairs only)			
madweight	0.778 $\pm$ 0.02	0.642 $\pm$ 0.03	0.354 $\pm$ 0.06
weight	0.777 $\pm$ 0.01	0.582 $\pm$ 0.02	0.408 $\pm$ 0.04
weight+TTT	0.777 $\pm$ 0.01	0.578 $\pm$ 0.02	0.411 $\pm$ 0.03
remove	0.755 $\pm$ 0.00	0.623 $\pm$ 0.02	0.411 $\pm$ 0.07
remove+TTT	0.748 $\pm$ 0.00	0.638 $\pm$ 0.02	0.388 $\pm$ 0.03
hybrid	0.760 $\pm$ 0.01	0.573 $\pm$ 0.02	0.399 $\pm$ 0.04
RESfM-scratch (sup.)	0.811 $\pm$ 0.01	0.623 $\pm$ 0.03	0.329 $\pm$ 0.06
ESfM (no mech.)	0.756 $\pm$ 0.01	0.570 $\pm$ 0.01	0.409 $\pm$ 0.04
ESfM* (clean tracks)	0.790 $\pm$ 0.05	0.669 $\pm$ 0.04	0.330 $\pm$ 0.07

Table 8. Per-scene win counts vs. the supervised baseline (matched-seed scene $\times$ seed cells, translation / rotation, as % of cells), and the catastrophic-cell census (cells with translation mean > 30 ; in parentheses, cells > 100). The adaptive label-free arms have zero > 100 cells anywhere; every > 100 cell in the matrix is an Olsson/dtu106-family failure of an unrescued-trunk recipe.

	trans %	rot %	trans/rot %	Cells > 100
supervised	48/68	0/5	50/55	42/47
Strech	48/56	15/40	55/55	45/50
BMVS	56/60	15/45	50/50	56/52
Olsson	80/84	0/0	30/50	39/42
1DSfM	72/80	0/0	55/60	36/42
hard	48/68	0/0	55/50	44/47
ESfM	44/36	15/40	38/38	55/51
ESfM*	52/52	15/35	38/25	45/39

Table 9. RA-AUC@30° vs. its Olsson score (0.620) sits above madweight (0.568): a 30°-capped pairwise metric dilutes a single catastrophic scene to its pair share, so the dtu106 failure that dominates the mean nearly vanishes from the AUC. The win-count and AUC lenses are therefore not more honest than means — they are differently blind: means amplify tails, capped counts hide them. The catastrophic-cell census (Table 8, right) is the tail-explicit summary: across $\sim 1,300$ scene $\times$ seed cells, the adaptive label-free arms produce zero cells above 100, while every unrescued-trunk recipe (supervised, weight+TTT, no-mechanism, oracle-clean) produces them, all on Olsson. Reliability — the absence of catastrophic cells — is the one verdict

both families so the reader can price coverage either way.

Counts, means, and tails disagree exactly where catastrophes live. ESfM* beats the supervised baseline on 58/59% of MegaDepth cells while losing the mean (0.594 vs. 0.365, medians 0.071 vs. 0.068) — clean tracks are typically better per scene and blow up occasionally. Symmetrically, the supervised baseline wins roughly half of the Olsson cells while losing the mean

587 on which all lenses agree.

588 The contamination axis under coverage pricing. Re-
589 scoring the in-distribution matrix (the main paper’s
590 in-distribution section) with RA-AUC@30° (five seeds
591 each): at 60%, supervision with best-case labels is
592 last on both metric families — worst-tier translation
593 mean (23.4 ± 7.8) and worst RA-AUC (0.012 ± 0.01 ,
594 from registering only ~ 106 of 600 cameras) — so
595 the causal label-ceiling verdict, unlike the OOD re-
596 moval ordering, is robust to how coverage is priced
597 (label-free soft arms lead at 0.088–0.090). At low and
598 moderate contamination the no-mechanism baseline’s
599 coverage advantage makes it the RA-AUC leader in-
600 domain (0.782/0.759/0.472 at 0.5/3.1/43%, vs. super-
601 vised 0.664 ± 0.27 — the wide band is its in-domain
602 bimodality); the mechanisms’ value under this lens, in
603 distribution, is concentrated exactly where labels fail.

604 L. Per-scene results and configuration

605 L.1. Per-scene results

606 Tables 9 and 10 report the per-scene post-BA transla-
607 tion and rotation errors and registered-camera counts
608 underlying the main paper’s dataset-level numbers
609 (seed 20, the declared pipeline): the 36 MegaDepth test
610 scenes and the ten 1DSfM scenes, for the three self-
611 supervised arms (madweight, weight, weight+TTT)
612 and the supervised RESfM baseline trained from
613 scratch under the identical recipe. The per-scene
614 view makes two aggregate statements concrete: on
615 1DSfM no single arm dominates, the best mech-
616 anism flips scene by scene (with *Ellis_Island* and
617 *Tower_of_London* hard for every arm), consistent
618 with the per-scene complementarity analysis in the
619 main paper; on MegaDepth the four arms are close
620 on most scenes and the dataset-level gap is dominated
621 by a handful of high-error scenes. Per scene, the
622 catastrophic scenes (*Ellis_Island*, *Tower_of_London*)
623 fail in both metrics, though rotation does not always
624 track translation (weight’s *Vienna_Cathedral* failure is
625 translation-specific: $22.71/0.46$), and N_r stays in the
626 58–92% range of each scene’s images on 1DSfM with
627 no arm systematically dropping scenes. The dataset-
628 level rotation and registration record accompanies the
629 main comparison (Table 2 of the main paper). Ro-
630 tation broadly agrees with translation at the dataset
631 level: the supervised BlendedMVS failure is even larger
632 in rotation (36°) and the high-contamination order-
633 ing is preserved, though the best arm can differ per
634 metric (on MegaDepth, weight+TTT leads transla-
635 tion while RESfM leads rotation). Registration is
636 contamination-dependent: the supervised baseline reg-

isters the most cameras in distribution (85% vs. 78–
80% on MegaDepth) and on clean Strecha, while un-
der high contamination the label-free arms retain more
cameras than supervision (38–43% vs. 37% on 1DSfM-
hard) at lower error (madweight).

642 M. Comparison with ESfM in both of its 643 regimes

ESfM is the no-outlier-mechanism ancestor of this
model family and operates in two regimes: multi-scene
training and its original per-scene optimization. We
compare in both.

Multi-scene (same contaminated tracks). Vanilla
ESfM — the ESfM architecture inside the RESfM
codebase, so trainer, tracks, and BA are identical to
every other arm — trained under the unified proto-
col (reference row in the main table; five-seed bands)
reaches 0.71 ± 0.11 on MegaDepth, 18.42 ± 0.75 on
1DSfM, 21.76 ± 0.80 at 60%, 2.03 ± 0.14 on Strecha,
 6.45 ± 3.75 on BlendedMVS, and 5.91 ± 3.32 on Olsson.
Three readings. First, without any outlier mechanism,
contaminated training data degrades every contami-
nated benchmark relative to all mechanism-equipped
arms — the premise of RESfM and of this paper. Sec-
ond, its apparent BlendedMVS competitiveness at seed
20 (0.23) dissolves at five seeds (6.45 ± 3.75 , bimodal)
— one more single-seed verdict reversed by banding
(cf. the seed-additivity analysis) — and its Olsson be-
haviour is likewise bimodal (collapse in three of five
seeds), so the clean-OOD failure mode predates the su-
pervised head and lives in the trunk. Third, on Strecha
the band (2.03 ± 0.14) statistically ties the best label-
free soft arms (1.97–2.00), confirming at band level that
at the clean extreme the correct mechanism approaches
none.

Single-scene (ESfM’s original per-scene optimization,
Olsson). Here the primary ESfM row runs in the au-
thors’ official ESfM codebase; the RESfM-codebase
reimplementation is included as a cross-check and
agrees to within noise (0.19 vs. 0.22°), ruling out
implementation gaps between the two regimes. On
36 calibrated Olsson scenes (3 runs each, matching
ESfM’s protocol), post-BA median-over-scenes rota-
tion/translation:

Table 9. Per-scene MegaDepth results (layout follows RESfM’s Table 1). For each scene we show the number of input images (N_c) and the fraction of outlier observations; for each model, the number of registered cameras (N_r) and mean rotation (e_R , degrees) and translation (e_t) errors, post-BA, seed 20. Above the middle rule are Group 1 scenes (< 1000 images); below are Group 2 scenes (> 1000 images, subsampled to 300 for testing), each group ordered by outlier fraction. Winning results per scene are marked in bold and underlined. weight+TTT values are single draws of a run-to-run nondeterministic evaluation (Sec. J.4).

Scene	N_c	%out	madweight			weight			weight+TTT			RESfM		
			N_r	e_R	e_t	N_r	e_R	e_t	N_r	e_R	e_t	N_r	e_R	e_t
0238	522	45%	437	19.02	1.41	137	8.23	0.62	141	8.37	0.63	355	<u>1.40</u>	<u>0.12</u>
0060	528	42%	469	0.29	0.03	476	0.20	0.02	452	4.13	0.37	494	<u>0.04</u>	<u>0.01</u>
0197	870	41%	470	7.22	0.89	341	3.54	0.38	422	6.94	1.11	739	<u>0.97</u>	<u>0.11</u>
0094	763	40%	493	2.72	0.45	505	1.44	<u>0.16</u>	432	<u>1.34</u>	0.17	572	5.33	0.88
0265	571	39%	457	17.62	3.26	477	5.78	2.34	547	<u>0.60</u>	<u>0.19</u>	424	16.27	3.09
0083	635	31%	342	4.34	0.23	511	17.70	1.56	317	3.20	0.20	582	<u>1.40</u>	<u>0.10</u>
0076	558	30%	423	1.40	0.38	519	0.61	0.23	454	0.74	0.22	499	<u>0.25</u>	<u>0.02</u>
0185	368	30%	347	0.21	0.01	335	0.24	0.04	321	0.88	0.09	352	<u>0.07</u>	<u>0.01</u>
0048	512	24%	475	1.93	0.14	467	<u>0.35</u>	<u>0.02</u>	450	0.43	0.03	489	1.85	0.11
0024	356	23%	299	1.27	0.18	291	<u>1.13</u>	<u>0.17</u>	289	4.78	1.13	334	3.67	1.09
0223	214	17%	205	2.66	0.24	198	1.69	0.23	199	3.06	0.30	204	<u>0.45</u>	<u>0.08</u>
5016	28	17%	23	0.27	0.04	25	0.08	0.02	28	<u>0.07</u>	0.02	28	0.08	<u>0.01</u>
0046	440	15%	422	<u>2.88</u>	<u>0.22</u>	421	4.11	0.30	419	3.88	0.25	410	14.64	0.93
0099	299	47%	224	<u>0.81</u>	0.15	216	2.77	3.52	212	1.58	<u>0.13</u>	252	2.18	0.39
1001	285	44%	280	1.58	1.22	278	1.49	0.73	279	<u>1.22</u>	<u>0.49</u>	273	1.23	0.50
0231	296	42%	274	1.63	0.15	257	0.64	0.07	264	1.49	0.24	265	<u>0.17</u>	<u>0.02</u>
0411	299	30%	281	0.22	0.03	245	2.99	0.32	250	4.12	0.46	275	<u>0.11</u>	<u>0.02</u>
0377	295	28%	200	0.62	0.05	222	2.76	0.26	243	<u>0.56</u>	<u>0.03</u>	223	0.82	0.06
0102	299	26%	289	0.15	0.02	289	<u>0.07</u>	<u>0.01</u>	291	0.08	0.01	287	0.10	0.02
0147	298	25%	261	31.69	2.99	235	15.28	1.21	225	15.27	<u>1.20</u>	247	<u>4.61</u>	3.22
0148	287	25%	220	2.60	0.33	201	2.11	0.11	196	<u>0.62</u>	0.04	202	0.94	<u>0.04</u>
0446	298	22%	292	<u>0.40</u>	<u>0.03</u>	285	1.20	0.19	292	1.04	0.19	293	0.71	0.05
0022	297	21%	275	0.14	0.02	272	0.22	0.03	267	0.33	0.04	276	<u>0.11</u>	<u>0.02</u>
0327	298	21%	282	1.97	0.09	285	0.22	0.02	284	0.21	0.02	277	<u>0.09</u>	<u>0.01</u>
0015	284	21%	196	2.64	0.30	196	3.87	0.56	211	3.90	0.65	232	<u>1.31</u>	<u>0.30</u>
0455	298	20%	283	<u>0.20</u>	<u>0.01</u>	285	0.21	0.01	288	0.46	1.60	293	0.85	0.09
0496	297	19%	279	<u>0.27</u>	0.04	285	0.50	<u>0.04</u>	287	1.28	0.09	283	0.46	0.07
1589	299	17%	265	0.96	0.06	262	0.28	0.05	296	1.91	0.30	291	<u>0.06</u>	<u>0.01</u>
0012	299	16%	281	<u>0.23</u>	<u>0.02</u>	247	9.51	0.70	262	2.76	0.22	288	5.53	0.34
0104	284	16%	189	0.32	0.03	193	2.28	0.18	194	1.44	0.07	199	<u>0.30</u>	<u>0.02</u>
0019	299	15%	232	3.22	0.17	253	6.55	0.28	264	3.24	0.13	252	<u>0.05</u>	<u>0.01</u>
0063	293	14%	258	0.63	0.08	268	0.72	0.07	267	0.65	<u>0.06</u>	266	<u>0.44</u>	0.07
0130	285	14%	250	10.17	1.60	184	1.87	0.16	185	<u>0.33</u>	<u>0.01</u>	195	1.03	0.07
0080	284	13%	132	1.10	<u>0.05</u>	148	2.48	0.20	147	0.70	0.07	141	<u>0.66</u>	0.10
0240	298	12%	275	<u>2.23</u>	<u>0.25</u>	275	4.64	0.34	272	5.32	0.46	282	3.02	0.35
0007	290	12%	106	<u>0.07</u>	<u>0.01</u>	269	1.86	0.09	261	3.28	0.14	104	0.99	0.04

Method (per-scene optimization)	Rot (°)	Trans
ESFM (official code)	0.19	0.057
ESFM (RESfM codebase)	0.22	0.088
ESFM + sequential init (10 scenes)	0.10	0.032
ESFM + MAD statistic	0.68	0.123
self-supervised arm (base)	8.03	0.748
self-supervised arm (30/70 percentiles)	2.81	0.434
self-supervised arm + TTT	1.22	0.324

Vanilla ESFM wins this regime outright, and every outlier mechanism — statistical or learned — costs accuracy. This is not a failure of the mechanisms but the low-contamination end of the complementarity finding replicated in a second regime: at 0.5% contamination with per-scene optimization there is nothing to reject,

and any removal or reweighting machinery only discards signal. The practical reading matches the main paper: mechanism choice must adapt to contamination, and “none” is the correct mechanism at the clean extreme of the axis.

N. Cross-generalization: every training domain, every test domain

We train both recipes on each of the six domains (in-distribution protocol of the main paper’s in-distribution study: GT-derived labels for the supervised arm) and evaluate every model on all datasets (~5,400 evaluations; five training seeds per cell,

Table 10. Per-scene 1DSfM results (layout follows RESfM’s Table 1). Columns as in Table 9; seed 20, scenes ordered by outlier fraction. Winning results per scene are marked in bold and underlined.

Scene	N_c	%out	madweight			weight			weight+TTT			RESfM		
			N_r	e_R	e_t	N_r	e_R	e_t	N_r	e_R	e_t	N_r	e_R	e_t
Ellis_Island	227	56%	180	17.16	17.59	171	17.37	15.90	177	17.74	17.11	192	18.15	16.02
NYC_Library	332	54%	282	2.32	2.94	261	1.16	1.19	279	10.30	3.39	285	12.99	6.28
Vienna_Cathedral	836	51%	537	7.42	7.30	484	0.46	22.71	651	26.80	33.57	492	1.09	2.29
Notre_Dame	553	48%	507	3.33	0.68	491	1.00	0.34	485	0.83	0.33	493	1.08	0.49
Yorkminster	437	45%	299	8.05	5.79	371	14.95	30.87	380	12.20	10.79	309	6.88	8.77
Piazza_del_Popolo	338	42%	218	0.43	0.94	265	19.54	10.66	254	20.46	17.05	255	3.15	2.49
Tower_of_London	472	42%	317	35.61	76.34	337	23.30	28.29	314	17.91	72.61	299	49.66	67.95
Montreal_Notre_Dame	450	36%	311	0.59	0.94	325	0.55	11.35	320	0.81	1.24	400	0.59	0.93
Alamo	577	29%	467	2.45	1.77	507	2.74	2.18	497	2.49	1.67	502	2.42	1.73
Madrid_Metropolis	341	29%	249	5.72	9.63	301	2.84	4.37	295	2.35	3.52	287	1.82	3.38

Table 11. Training and evaluation configuration. All arms in the main paper use these settings; the only per-method difference is the base learning rate.

Setting	Value
Training (both methods)	
Epochs	20,000 (one epoch = one pass over all scenes)
Batch	4 scenes
LR schedule	milestones 10k, 11k, ..., 20k, $\gamma = 0.9$
Base LR	10^{-4} (SS-RESfM) / 10^{-3} (RESfM)
Loss weights	$\alpha_{\text{reproj}} = 1$, $\beta_{\text{cls}} = 1$
Infinity-point margin	10^{-4}
Checkpoint selection	best validation error, evaluated every 500 epochs
Seed	20 (21–22 / 21–24 for the seed studies)
Adaptive pseudo-labels (SS-RESfM only)	
Inlier / outlier percentile	20 / 80 (of per-observation reproj. error)
Threshold separation	scale-aware floor, $0.25 \times$ median error
Warm-up	10 epochs before pseudo-labeling starts
Evaluation	
Removal threshold τ	0.6 (0.8 for low-contamination single-camera sets, following RESfM)
Test-time fine-tuning	1,001 iterations, LR 5×10^{-3} , evaluated every 250 iterations
Bundle adjustment	pyceres/pycolmap robust BA, as RESfM

in all 27 cells; excluding dtu106, a modest supervised deficit persists in four of five pools (3.4–4.3 vs. 2.7–3.3). The reliability asymmetry — both the lottery and the residual gap — is a property of the recipes, not of MegaDepth. (ii) The Strecha exception is likewise invariant: supervision wins Strecha from every training domain (0.72–2.37 vs. 2.56–3.05). (iii) At the extreme (60% target), madweight from any domain beats supervision from any domain (best supervised cell 18.59 vs. madweight 15.56–19.70, with the in-distribution madweight at 12.1) — and, remarkably, MegaDepth as a test set is nearly training-domain-independent (all cells 0.47–0.83), suggesting its curated test scenes are easy for any reasonably trained model of this family. One counterexample sharpens the picture: supervised trained on 1DSfM-hard generalizes to standard 1DSfM better than anything else (8.49), consistent with supervision learning transferable contamination statistics when its labels are sound on a related domain.

O. Per-scene label degradation

Table 13 gives the per-scene data behind the main paper’s label-ceiling analysis: for each of the 18 scenes we measure the contamination level, the fraction of track observations that receive a COLMAP-derived label at all (coverage), and the precision and recall of those labels against ground truth. The aggregate trend in the main paper is visible scene by scene: below $\sim 10\%$ contamination coverage is 0.92–1.00 and recall is 0.66–0.97 (mostly above 0.8), while above $\sim 15\%$ contamination recall never exceeds 0.50 on any scene (0.25–0.50) and coverage falls as low as 0.37: the supervision signal degrades exactly where outlier handling matters most.

P. Pre- versus post-BA seed variability

Table 14 reports, for every arm and dataset, the five-seed band of the network output (pre-BA) next to the

strechaid two). Cells where a pool’s training scenes overlap the target are excluded. Table 12 reports madweight / supervised post-BA translation means.

Three findings generalize beyond MegaDepth training. (i) The clean-benchmark collapse is training-domain-invariant — and so is its tail structure: the supervised recipe degrades on Olsson OOD from every training domain (4.96–8.62 across five pools, seed-volatile up to ± 4.25), while madweight lands in a tight 2.69–3.34 band from every pool. The tail decomposition of the main paper holds pool-by-pool: the supervised dtu106 catastrophic failure fires from every training domain (on 1–5 of 5 seeds per pool, errors ~ 230 –360 vs. ~ 1 on the escaping seeds — a pure basin lottery), whereas madweight stays at ≤ 7 on that scene

Table 12. Cross-generalization grid: rows = training domain, columns = test dataset; each cell madweight / supervised (mean $\pm$ std over training seeds). Diagonal (*) = held-out in-distribution test (Table 4 of the main paper); “leak” = training scenes overlap the target.

Trained on $\downarrow$	MegaDepth	Olsson	Strecha	BMVS	1DSfM	1DSfM-hard
Olsson	0.52 / 0.62	*1.97 / 2.03	3.05 / 0.72	6.97 / 6.70	11.84 / 12.04	15.56 / 20.17
Strecha (2 seeds)	0.53 / 0.54	2.69 / 8.62	*0.01 / 0.01	7.29 / 8.08	13.94 / 10.22	17.06 / 18.97
BMVS	0.63 / 0.51	2.73 / 6.95	2.56 / 1.44	*0.09 / 0.26	11.50 / 10.45	19.70 / 18.59
1DSfM-easy	0.53 / 0.56	3.27 / 7.20	2.94 / 1.92	6.48 / 5.81	*6.37 / 4.97	18.31 / 20.12
1DSfM-hard	0.47 / 0.83	2.93 / 7.78	3.02 / 1.94	3.65 / 5.68	13.21 / 8.49	*12.1 / 23.4
1DSfM-merged	0.52 / 0.59	3.34 / 4.96	2.98 / 2.37	7.26 / 6.69	*6.71 / 5.05	leak

Table 13. Per-scene label-degradation results. For each scene: contamination (fraction of track observations that are GT outliers), COLMAP-label coverage (fraction of observations receiving a label), and precision/recall of the COLMAP-derived outlier labels against GT, sorted by contamination. Above $\sim 15\%$ contamination, label recall never exceeds 0.50: the labels miss at least half of the true outliers.

Dataset	Scene	Contam. (%)	Coverage	Prec.	Recall
BlendedMVS	58c4bb4f4a...	0.5	1.00	0.86	0.94
Strecha	Herz-Jesu-P8	0.7	1.00	0.51	0.56
Strecha	fountain-P11	0.7	1.00	0.69	0.50
Strecha	Herz-Jesu-P25	2.6	0.99	0.72	0.82
Strecha	entry-P10	2.7	0.99	0.88	0.88
BlendedMVS	5be883a4f9...	4.0	0.98	0.90	0.84
BlendedMVS	5b6e716d67...	6.1	0.92	0.89	0.84
BlendedMVS	5bfd0f32ec...	7.1	0.97	0.97	0.90
BlendedMVS	5b950c7160...	9.2	0.96	0.92	0.94
MegaDepth	0007	11.2	0.73	0.84	0.47
1DSfM	Alamo	26.6	0.76	0.81	0.25
1DSfM	Madrid_Metropolis	28.8	0.67	0.85	0.25
MegaDepth	0041	31.4	0.49	0.80	0.43
MegaDepth	0060	39.0	0.42	0.53	0.45
1DSfM	Piazza_del_Popolo	41.9	0.76	0.90	0.30
MegaDepth	0047	54.9	0.37	0.83	0.30
1DSfM	Ellis_Island	57.9	0.79	0.89	0.30
1DSfM-hard	Gendarmenmarkt	59.8	0.50	0.89	0.29

tween independently trained networks, amplified by BA, and not evaluation noise. A full paired-seed replication confirms this at scale: re-running all 2,352 evaluations with the evaluation seed set equal to each cell’s training seed (rather than fixed at 20) reproduces every five-seed band within a fraction of its std — e.g. the baseline row moves from 0.37 ± 0.12 to 0.38 ± 0.14 on MegaDepth and is unchanged to two decimals on Strecha, BlendedMVS, and Olsson — with the largest shifts in the TTT cells (~ 0.13), consistent with their row-to-row nondeterminism. The choice of evaluation seed is immaterial to every conclusion in this paper.

First, on the contaminated datasets nearly all seed variability is created by bundle adjustment, not by training: pre-BA bands are ± 0.02 – 0.13 for the self-supervised arms and ± 0.05 – 0.43 for the baseline (relative spreads of $\sim 0.5\%$ and 1 – 3% respectively), while the same cells after BA carry ± 1.7 – 4.1 . The networks are highly reproducible across seeds; most of the pre-BA’s variance enters at the BA stage, which lowers the error on 234 of 242 baseline scene evaluations we examined while widening the relative seed band by an order of magnitude — BA compresses the mean and amplifies the spread. This is the seed-level counterpart of the environment-level chaotic basin selection documented in App. J, now measured over 36 independent cells.

Second, the removal arms do not produce better raw geometry than supervision — they produce geometry that robust BA can repair. Pre-BA the supervised baseline leads everywhere (MegaDepth 1.76 vs. ~ 3.03 ; 1DSfM 22.1 vs. 25.3–26.6), yet plain removal improves by $2.9\times$ through BA at 60% contamination ($36.3 \rightarrow 12.4$) against the baseline’s $1.9\times$ ($35.8 \rightarrow 19.2$), and by $5.0\times$ on Olsson ($17.1 \rightarrow 3.4$) against $1.5\times$ ($11.0 \rightarrow 7.4$). Feeding BA a cleaner observation set, rather than a more accurate initialisation, is the mechanism behind the high-contamination result.

A third observation concerns the baseline specifi-

band after robust bundle adjustment. The behaviour is what one expects of a non-linear optimiser with multiple basins, and it is worth recording because two statements in the main text depend on it.

A direct control separates the two sources. Re-evaluating the authors’ fixed released weights under five evaluation seeds gives 0.203 ± 0.008 on MegaDepth (pre-BA 1.730 ± 0.005), 10.71 ± 0.27 on 1DSfM, and identical values to two decimals on Strecha (0.20) and BlendedMVS (0.33): with the network held constant, bundle adjustment is close to deterministic, and exactly deterministic on the small clean sets. The large bands below therefore reflect genuine differences be-

801 cally, and is the one exception to the first: its clean-
802 OOD instability is already present before BA (Strecha
803 0.68 ± 0.78 , Olsson 11.01 ± 2.27 , versus ± 0.18 – 0.24
804 and ± 0.19 – 0.47 for the self-supervised arms), locating
805 that failure in supervised training rather than in bundle
806 adjustment or evaluation. Per scene the failure is
807 bimodal and seed-contingent rather than noisy: entry-
808 P10 costs 0.04 – 0.06 on three seeds and $4.0/7.0$ on the
809 other two; dtu106 sits at 250 – 260 on four seeds and 53
810 on one. Five authors-faithful retrains (seeds 20 – 24 , the
811 authors’ validation metric and margin) reproduce the
812 scratch baseline’s band on every dataset — including
813 the one-bad-Strecha-seed pattern and the four-of-five
814 Olsson collapse (8.66 – 9.02 with a single 2.32 escape,
815 mirroring the scratch 2.69) — so these failure modes
816 belong to the supervised recipe, not to our retraining
817 of it. The same signature appears in-distribution
818 at smaller amplitude — individual MegaDepth scenes
819 swing 3 – $4\times$ across seeds (0060: 1.0 – 3.3 ; 0231: 0.5 –
820 2.2) while opposite-signed swings cancel in the dataset
821 mean, and the baseline’s median per-scene std is about
822 twice the arms’ (0.11 vs. 0.04 – 0.05). Our reading is
823 structural: the supervised head fits fixed labels gener-
824 ated at $\sim 25\%$ contamination, many decision bound-
825 aries achieve near-identical training loss, and its be-
826 haviour on scenes far from that regime — near-outlier-
827 free ones in particular — is underdetermined and de-
828 cided by the draw of the seed. The self-supervised
829 arms, trained against their own reprojection-error per-
830 centiles, converge to nearly the same solution from
831 every seed (per-scene stds as low as 0.01). Stability
832 is not accuracy — the arms remain worse pre-BA on
833 clean data — but predictability under retraining is a
834 property the label-free objective buys that labels do
835 not. The single-run reference rows in the lower block
836 sharpen the same point: every supervised variant, in-
837 cluding the authors’ released weights, leaves BA with
838 an Olsson error of 8.1 – 9.1 despite entering it at 11.9 –
839 12.1 — a repair factor of only $\sim 1.4\times$, against 4.7 – $5.5\times$
840 for the label-free arms from a worse starting point
841 (13.4 – $17.1 \rightarrow 2.9$ – 3.6). On the clean benchmark the
842 supervised head does not give bundle adjustment a set
843 of observations it can fix.

Table 14. Pre-BA $\rightarrow$ post-BA translation error, mean $\pm$ std over five training seeds (seeds 20–24). Pre-BA bands are one to two orders of magnitude tighter than post-BA bands, except the supervised baseline’s clean-OOD cells (Strecha, Olsson), whose variance predates BA (see text). [§]For the released checkpoint the MegaDepth entry is a band over five evaluation seeds on the fixed weights (± 0.008), for comparison with the training-seed bands above: re-seeding the evaluation moves the result by two orders of magnitude less than retraining does.

Arm	MegaDepth	1DSfM	1DSfM-hard	Strecha	BlendedMVS	Olsson
madweight	3.03 $\pm$ 0.02 $\rightarrow$ 0.51 $\pm$ 0.08	26.14 $\pm$ 0.04 $\rightarrow$ 11.04 $\pm$ 1.68	36.33 $\pm$ 0.07 $\rightarrow$ 18.92 $\pm$ 4.05	3.77 $\pm$ 0.24 $\rightarrow$ 2.65 $\pm$ 0.25	0.58 $\pm$ 0.03 $\rightarrow$ 0.14 $\pm$ 0.02	16.38 $\pm$ 0.45 $\rightarrow$ 3.01 $\pm$ 0.34
weight	3.04 $\pm$ 0.03 $\rightarrow$ 0.51 $\pm$ 0.09	26.60 $\pm$ 0.09 $\rightarrow$ 14.35 $\pm$ 2.97	36.26 $\pm$ 0.04 $\rightarrow$ 22.92 $\pm$ 2.79	2.57 $\pm$ 0.19 $\rightarrow$ 1.97 $\pm$ 0.09	0.51 $\pm$ 0.01 $\rightarrow$ 0.14 $\pm$ 0.03	15.86 $\pm$ 0.28 $\rightarrow$ 2.91 $\pm$ 0.33
weight+TTT	3.01 $\pm$ 0.03 $\rightarrow$ 0.40 $\pm$ 0.11	26.63 $\pm$ 0.10 $\rightarrow$ 15.72 $\pm$ 2.10	36.20 $\pm$ 0.12 $\rightarrow$ 22.24 $\pm$ 3.17	2.63 $\pm$ 0.21 $\rightarrow$ 2.00 $\pm$ 0.05	0.50 $\pm$ 0.02 $\rightarrow$ 0.14 $\pm$ 0.02	13.41 $\pm$ 0.19 $\rightarrow$ 8.03 $\pm$ 0.30
remove	3.03 $\pm$ 0.02 $\rightarrow$ 0.51 $\pm$ 0.08	25.34 $\pm$ 0.10 $\rightarrow$ 9.13 $\pm$ 2.17	36.32 $\pm$ 0.09 $\rightarrow$ 12.42 $\pm$ 2.05	4.22 $\pm$ 0.19 $\rightarrow$ 3.20 $\pm$ 0.11	0.62 $\pm$ 0.01 $\rightarrow$ 0.24 $\pm$ 0.06	17.05 $\pm$ 0.39 $\rightarrow$ 3.41 $\pm$ 0.57
remove+TTT	3.04 $\pm$ 0.02 $\rightarrow$ 0.56 $\pm$ 0.07	25.23 $\pm$ 0.13 $\rightarrow$ 8.93 $\pm$ 1.20	36.32 $\pm$ 0.07 $\rightarrow$ 15.15 $\pm$ 0.98	4.31 $\pm$ 0.18 $\rightarrow$ 3.01 $\pm$ 0.07	0.63 $\pm$ 0.02 $\rightarrow$ 0.22 $\pm$ 0.10	17.03 $\pm$ 0.47 $\rightarrow$ 3.63 $\pm$ 0.35
RESfM-scratch	1.76 $\pm$ 0.05 $\rightarrow$ 0.37 $\pm$ 0.12	22.13 $\pm$ 0.27 $\rightarrow$ 10.50 $\pm$ 1.45	35.80 $\pm$ 0.43 $\rightarrow$ 19.16 $\pm$ 3.42	0.68 $\pm$ 0.78 $\rightarrow$ 0.39 $\pm$ 0.72	0.45 $\pm$ 0.01 $\rightarrow$ 0.35 $\pm$ 0.04	11.01 $\pm$ 2.27 $\rightarrow$ 7.37 $\pm$ 2.63
RESfM-faithful	1.80 $\pm$ 0.09 $\rightarrow$ 0.32 $\pm$ 0.11	22.20 $\pm$ 0.16 $\rightarrow$ 11.12 $\pm$ 1.54	35.74 $\pm$ 0.22 $\rightarrow$ 17.93 $\pm$ 3.64	0.64 $\pm$ 0.65 $\rightarrow$ 0.38 $\pm$ 0.66	0.46 $\pm$ 0.01 $\rightarrow$ 0.35 $\pm$ 0.03	11.12 $\pm$ 2.66 $\rightarrow$ 7.53 $\pm$ 2.92
single-run reference rows (no seed band; pre $\rightarrow$ post only)						
RESfM-faithful @10 ⁻⁴	2.23 $\rightarrow$ 0.60	22.88 $\rightarrow$ 6.72	34.98 $\rightarrow$ 16.21	1.27 $\rightarrow$ 0.22	0.32 $\rightarrow$ 0.17	12.00 $\rightarrow$ 8.11
RESfM-released [§]	1.730 $\pm$ 0.005 $\rightarrow$ 0.203 $\pm$ 0.008	22.42 $\rightarrow$ 10.71 $\pm$ 0.27	35.26 $\rightarrow$ 15.29	0.97 $\rightarrow$ 0.20 $\pm$ 0.00	0.42 $\rightarrow$ 0.33 $\pm$ 0.00	12.11 $\rightarrow$ 9.10
RESfM @10 ⁻⁴ , 30k	2.08 $\rightarrow$ 0.36	23.14 $\rightarrow$ 6.12	35.76 $\rightarrow$ 13.88	1.92 $\rightarrow$ 1.64	0.29 $\rightarrow$ 0.18	11.94 $\rightarrow$ 8.27